

DTSA 5502

Trees and Graphs: Basics

|                                                                                 |                          |               |
|---------------------------------------------------------------------------------|--------------------------|---------------|
| FORMAT                                                                          | CATEGORY                 | CREDITS       |
| Online (DTSA)                                                                   | Core                     | 1             |
| SECTION                                                                         | BEST ROLE MATCH          | AVG ALIGNMENT |
| Section 1 – CS Pathway                                                          | AI Engineer (27.5 / 100) | 17.8 / 100    |
| SPECIALIZATION Data Science Foundations: Data Structures and Algorithms Pathway |                          |               |

18

CURRENT AVG SCORE  
across 6 job roles

65

POTENTIAL SCORE  
with recommended additions

+68

MAX SCORE GAIN  
best single-role improvement

CURRENT COURSE DESCRIPTION

Basic algorithms on tree data structures, binary search trees, self-balancing trees, graph data structures and basic traversal algorithms on graphs. This course also covers advanced topics such as kd-trees for spatial data and algorithms for spatial data.

INDUSTRY ALIGNMENT BY JOB ROLE

| JOB ROLE                  | CURRENT SCORE               | PROJECTED SCORE             | POINT GAIN | PRIORITY |
|---------------------------|-----------------------------|-----------------------------|------------|----------|
| AI Engineer               | <div><div></div></div> 27.5 | <div><div></div></div> 61.2 | +33.7      | HIGH     |
| ML Engineer               | <div><div></div></div> 21.8 | <div><div></div></div> 56.7 | +34.9      | HIGH     |
| Data Scientist            | <div><div></div></div> 19.9 | <div><div></div></div> 87.4 | +67.5      | HIGH     |
| Machine Learning Engineer | <div><div></div></div> 14.9 | <div><div></div></div> 63.2 | +48.4      | HIGH     |
| Applied Scientist         | <div><div></div></div> 11.8 | <div><div></div></div> 74.3 | +62.6      | HIGH     |
| MLOps Engineer            | <div><div></div></div> 10.8 | <div><div></div></div> 48.7 | +37.9      | HIGH     |

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold ( example ) appear as gaps across 3 or more job roles.

System Design

4 of 6 roles

systems

Business Communication

1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

AI Engineer

27.5 → 61.2 (+33.7 pts)

following models learning software systems problems detection machine

ML Engineer

21.8 → 56.7 (+34.9 pts)

discovery learning machine systems models world problems product

Data Scientist

19.9 → 87.4 (+67.5 pts)

sets product analytics different social problems

Machine Learning Engineer

14.9 → 63.2 (+48.4 pts)

sets learning analytics tools solutions business mobile design

Applied Scientist

11.8 → 74.3 (+62.6 pts)

discovery scientific research interdisciplinary problems models learning systems

MLOps Engineer

10.8 → 48.7 (+37.9 pts)

detection systems learning machine tools models development error

**Methodology:** Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.